

Completely randomised design (CRD)

- To compare 'h' treatments
- The plots are numbered from 1 to n serially.
- Randomly selecting samples and give the treatment.

Randomized Block Design (RBD)

Two way classification

- To test the effect of k fertilizing treatments on the yield of a crop
- Divide the plots in h blocks, each block containing k plots
- Within each block, the k treatments are given to the k plots in a perfectly random manner, such that each treatment occurs only once in any block.
- But the same k treatments are repeated from block to block.

Analysis of variance (ANOVA)

VARIATION	SUM OF SQUARES	DEGREE OF FREEDOM	MEAN SQUARE	VARIANCE RATIO (F TEST)
BETWEEN TREATMENTS	Q1	h-1	$S1^2 = Q1 / (h-1)$	Calculated value of $F = S1^2 / S2^2$
WITHIN TREATMENTS	Q2	N-h	$S2^2 = Q2 / (N-h)$	Table value of with LOS $F(h-1, N-h)$
TOTAL	Q	N-1		

Formulae for finding Q1, Q2 and Q

$$Q = \Sigma \Sigma x_{ij}^2 - \frac{T^2}{N}$$

Correction factor T is the total value of observations

$$Q1 = \Sigma \frac{T_i^2}{n_i} - \frac{T^2}{N}$$

$$Q2 = Q - Q1$$

Working procedure

H0: There is no significant difference between column means

H1: There is significant difference between column means

Step 1: Find N , the number of observations

Step 2: Find T , the total value of all observations

Step 3: Find T^2/N , the correction factor

Step 4: calculate the total sum of squares Q

$$Q = \Sigma X_1^2 + \Sigma X_2^2 + \dots - \frac{T^2}{N}$$

Step 5: Calculate the Q1 (sum of squares between treatments)

$$Q1 = \frac{(\Sigma X_1)^2}{N_1} + \frac{(\Sigma X_2)^2}{N_2} + \dots - \frac{T^2}{N}$$

Step 6: Prepare the ANOVA table to calculate F-ratio

Step 7: Find the table value.

Step 8: Conclusion

Problems

A test was given to 5 students chosen at random from B.Tech., class of each of the three universities in Chennai. Their scores were found as follows: Test the significant difference between three universities.

Universi ty	Scores				
A	9	7	6	5	8
B	7	4	5	4	5
C	6	5	6	7	6

Problem-2

A car rental agency, which uses 5 different brands of tyres in the process of deciding the brand of tyre to purchase as standard equipment for its fleet, finds that each of 5 tyres of each brand last the following number of kilometres (in thousands):

A	B	C	D	E
36	46	35	45	41
37	39	42	36	39
42	35	37	39	37
38	37	43	35	35
47	43	38	32	38

Test the hypothesis that the five tyre brands have almost the same average life.

The following figures relate to production in kgs. of three variables A,B,C of wheat sown on 12 plots.

A	14	16	18		
B	14	13	15	22	
C	18	16	19	19	20

Is there any significant difference in the production of the variables.

ANOVA for RBD (two way)

VARIATION	SUM OF SQUARES	DEGREE OF FREEDOM	MEAN SQUARE	VARIANCE RATIO (F TEST)	TABLE VALUE
BETWEEN THE COLUMNS	QC	C-1	$SC^2 = QC / (C-1)$	Calculated value of $FC = SE^2 / SC^2$	$FC(N-C-R+1, C-1)$
BETWEEN THE ROWS	QR	R-1	$SR^2 = QR / (R-1)$	Calculated value of $FR = SE^2 / SR^2$	$FR(N-C-R+1, R-1)$
ERROR	QE	N-C-R+1	$SE^2 = QE / (N-C-R+1)$		
TOTAL	Q	N-1			

Formula for RBD

calculate the total sum of squares Q

$$Q = \Sigma X_1^2 + \Sigma X_2^2 + \dots - \frac{T^2}{N}$$

$$QC = \frac{(\Sigma X_1)^2}{N_1} + \frac{(\Sigma X_2)^2}{N_1} + \dots - \frac{T^2}{N}$$

$$QR = \frac{(\Sigma Y_1)^2}{N_2} + \frac{(\Sigma Y_2)^2}{N_2} + \dots - \frac{T^2}{N}$$

$$QE = Q - QC - QR$$

Example

The following table gives monthly sales (in thousand rupees) of a certain firm in the 3 states by its four salesmen.

States	Salesmen			
	I	II	III	IV
A	6	5	3	8
B	8	9	6	5
C	10	7	8	7

Setup the analysis of variance table and test whether there is any significant difference (i) between the salesmen (ii) between sales in the states.

Example

Perform a two-way ANOVA on the data given below.

Treatment 1	Treatment 2		
	A	B	C
A	30	26	38
B	24	29	28
C	33	24	35
D	36	31	30
E	27	35	33

ANOVA for LSD (Three way)

VARIATION	SUM OF SQUARE S	DEGREE OF FREEDOM	MEAN SQUARE	VARIANCE RATIO (F TEST)	TABLE VALUE
BETWEEN THE COLUMNS	QC	k-1	$SC^2 = QC / (k-1)$	$FC = SE^2 / SC^2$	$F((k-1)(k-2), (k-1))$
BETWEEN THE ROWS	QR	k-1	$SR^2 = QR / (k-1)$	$FR = SE^2 / SR^2$	$F((k-1)(k-2), (k-1))$
BETWEEN THE TREATMENTS	QT	k-1	$ST^2 = QT / (k-1)$	$FT = SE^2 / ST^2$	$F((k-1)(k-2), (k-1))$
ERROR	QE	$(k-1)(k-2)$	$SE^2 = QE / (k-1)(k-2)$		
TOTAL	Q	N-1			

Example

The following is a LSD when 4 varieties of seeds are being tested. Set up the ANOVA and state your conclusion.

A105	B95	C125	D115
C115	D125	A105	B105
D115	C95	B105	A115
B95	A135	D95	C115

Example

Study the effectiveness of three teaching methods A,B and C from the achievement scores given below tabulated age and aptitude wise.

aptitude	Age		
	Young	Middle	Old
Low	A82	B87	C80
Medium	B92	C82	A81
High	C90	A83	B88